STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 1: Units and Measurements

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. How many SI base quantities? [1]
2. SI unit of temperature? [1]
3. Significant figures in 0.00450? [1]
4. Dimensions of speed? [1]
5. Dimensions of force? [1]
6. Is length/length dimensionless? [1]
7. Can dimensional analysis determine pi? [1]
8. SI unit of pressure? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. Round 12.486 to 3 significant figures. [2]
10. Dimensions of momentum mv? [2]
11. Check s=ut+at^2 dimensionally. [2]
12. Convert 72 km/h to m/s. [2]
13. Significant figures in 5.070x10^3? [2]
14. Why can’t dimensions distinguish work and torque? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. Derive pendulum period dependence T proportional to l^ag^b. [3]
16. Sphere radius 2.00 cm: volume to 3 s.f.? [3]
17. Dimensions of G from F=Gm_1m_2/r^2? [3]
18. 15.2 cm x 3.4 cm: area with correct precision? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Spring-mass period T proportional to m^ak^b, [k]=MT^-2. Find exponents. [4]
20. Calculated volume is 18.237 cm^3, least precise input 3 s.f. What is reported? [4]

End of paper - check working, units, signs and significant figures.